

Hemal Patel

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PROFESSIONAL SUMMARY

Senior Engineer with 10+ years shipping gameplay systems, multiplayer networking, and server-side infrastructure in Unreal Engine 5 and Unity. Diagnosed and resolved cross-platform build failures between Windows and Linux — including strict type enforcement, path separator incompatibilities, and OS-level threading divergence. Operated and debugged live Linux server deployments under broadcast constraints, validated dedicated server builds via WSL and Docker containers, and engineered CI/CD tooling adopted studio-wide. Comfortable owning platform-specific work end-to-end, from identifying root cause to shipping a stable cross-platform build.

SKILLS

Languages: C++, C#, Lua, Python, Bash

Engines & Frameworks: Unreal Engine 4/5 (Dedicated Server, Replication, GAS, Niagara, Unreal Insights, UBT, BuildGraph), Unity, Godot

Server & Platform: Cross-Platform Development (Windows/Linux), Dedicated Game Servers, Server-Authoritative Architecture, Performance Profiling & Optimization, Multithreading, Tick Stability

Tools & Pipeline: Unreal Build Tool, BuildGraph, Docker, WSL, CI/CD, Custom PAK Pipelines, RenderDoc, Git, Perforce, JIRA

Online & Backend: PlayFab (Matchmaking, Session Management, Inventory), NATS Streaming, REST APIs, AWS, Azure, Grafana, Analytics Pipelines

SHIPPED TITLES

[Wave VR](#) (Unity) | [Wave Beta](#) (Unreal) | [Wave Standalone](#) (Unreal) | [Alimentary Avengers](#) (Unity) | [Be Safe, Be Smart!](#)

Notable Experiences: The Weeknd | Justin Bieber | Calvin Harris | John Legend | Tinashe | Pentakill (League of Legends)

EXPERIENCE

Software Engineer II

Nov 2019 – Dec 2025

WaveXR

Remote (Horsham, PA)

- **Cross-Platform Build Ownership:** Diagnosed and resolved Windows-to-Linux build failures including strict type enforcement differences, path separator incompatibilities (resolved using Unreal platform-agnostic path templates), and threading race conditions caused by OS-level scheduler divergence — maintaining a stable, reproducible cross-platform build.
- **Linux Server Operations:** Operated Linux server instances during live concert broadcasts — pulling Grafana logs, configuring server-side services, and resolving runtime failures directly on Linux hosts under hard broadcast deadlines with zero tolerance for downtime.
- **Dedicated Server Validation:** Validated dedicated server builds using WSL and Docker containers, reproducing Linux-specific runtime behavior and confirming server stability prior to cloud deployment.
- **Multiplayer Networking Architecture:** Solo-architected a production NATS + Unreal networking system with Manager/Controller/UI pipelines, routing high-frequency performance data server-side, offloading Unreal replication, and enabling multi-instance distribution for live stage events.
- **Backend & Session Management:** Integrated PlayFab matchmaking, inventory, and player login systems, managing full session lifecycle and player data flows across client and server boundaries.
- **Build System & CI/CD:** Built a UE5 build-config tool (UBT XML) and custom PAK packager automating credential injection across 15 environments; containerized supporting services (NATS, build runners) with Docker and integrated the toolchain into CI/CD, adopted studio-wide.
- **Server Performance Optimization:** Rebuilt the concurrent Niagara particle system using the command pattern, boosting frame rate from 15 fps to 36 fps under full audience load; profiled and resolved RPC/Blueprint bottlenecks via Unreal Insights, cutting frame time and memory usage by over 80%.
- Led 2 engineers and 2 tech artists delivering the Social Experience feature from prototype to production, owning architecture, planning, and full lifecycle execution.
- Drove Unity to UE5 engine migration, architecting core systems in C++ and maintaining feature parity across the full product stack.
- Built a UE5 Segment analytics subsystem batching up to 20 timestamped REST events before dispatch, callable from any game system.

- Engineered interactive systems powering 6 live mocap concert experiences, each with unique production requirements and hard broadcast deadlines across Unity and Unreal Engine.

Unity Programmer (Contract)

Nov 2015 – Nov 2019

Razor Edge Games

Remote (Horsham, PA)

- Engineered core gameplay systems in C#, including turn-based simulation, inventory management, and survival mechanics, architecting clean, extensible code for long-term maintainability.
- Architected and implemented the Character Creator system in Unity from concept to completion, handling complex mesh, animation, and UI interactions.
- Built custom editor tools and extensions that streamlined the content creation pipeline, reducing iteration time for artists and designers.
- Established coding standards and led code reviews to ensure quality and maintainability across a distributed team.

Unity Developer

Jun 2014 – March 2015

Smartcloud Infotech Pvt. Ltd.

Pune, India

- Developed and shipped multiple interactive commercial applications on mobile platforms using Unity, delivering on strict client requirements and timelines.

TECHNICAL PROJECTS

UniversalDataSync | C++, Unreal Engine, Multiplayer Networking

- Architected an Unreal Engine plugin to streamline replication of arbitrary data structures across multiplayer sessions, reducing networking boilerplate for complex gameplay states.

NeoThreads | C++, Unreal Engine, Multithreading

- Developed a lightweight concurrency framework offloading CPU-intensive operations from the Game Thread, improving frame stability and tick consistency for server-side gameplay systems.

ElementalArsenal | C++, Unreal Engine 5, GAS

- Engineered a modular GAS-based gameplay plugin supporting data-driven equipment and automated elemental reactions, built with portability and server-authoritative design in mind.

EDUCATION

DePaul University

MS in Computer Game Programming

Online
In Progress

CEPT University

Master of Business Administration, Information Technology and Project Management

India

2014

Gujarat Technological University

Bachelor of Technology in Information Technology

India

2012